

Comparative analysis of SOC estimation models

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Abstract—The rapid growth of electric vehicles underscores the critical need for the efficient battery management, with accurate State of Charge(SOC) estimation, which optimizes battery performance and life. Traditional SOC estimation methods like electrical and electrochemical equivalent circuit methods rely on complex mathematical PDEs and empirical techniques which increase computational cost, and also face adaptability challenges. Thus, addressing such challenges, a data-driven method is used in this project to estimate the SOC by learning the model from the given battery dataset. This study explores data-driven approaches using CNN, LSTM, and hybrid CNN-LSTM models to estimate SOC and give the comparative performance analysis of these models. The dataset used in this project is a labeled dataset of an EV which contains current, voltage, timestamp, and SOC. Upon exploring the models and tuning the hyperparameters LSTM performed best for sequential data with 80% accuracy and CNN performed worst.

I. INTRODUCTION

The global transition towards electric vehicles is gaining momentum due to environmental consciousness and also due to technological advancement. According to the recent trends, electric vehicle sales will make a considerable portion of the total automotive market share in years to come. So, all the concerns are focused on making the battery and its management system efficient and reliable which can optimize the vehicle's performance. Thus, accurate estimation of SOC is most crucial to increase battery efficiency. State of charge indicates the remaining charge in the battery at the given point of time. Accurate estimation of SOC is essential to enhance efficiency, increase safety and life of the battery [1].

$$\text{SOC} = \frac{q(t)}{q}$$

SOC strongly depends on the measurable battery parameters such as voltage, current, temperature and these variables are commonly used for estimation. In recent years, study has been diverted to data-driven approaches where Machine Learning algorithms are used to model the battery from the collected data at different conditions. Traditional, Equivalent Circuit-based Models, empirical models and electrochemical models estimating SOC levels directly have high computational complexities and are not cost effective. Also, ML methods, including Kalman filters, neural networks, fuzzy logic controllers and various hybrid techniques, have been extensively studied in the literature. Data-driven approaches leverage the powerful nonlinear mapping capabilities of

machine learning and aim to map input data to outputs[2]. In the past few years, data-driven methods have been widely used for EOD prediction and aging inference and there is a growing trend toward leveraging deep learning techniques for battery modeling. Deep Learning based approaches, such as LSTM, GRU, DNN, and CNN-LSTM, have demonstrated promising results. Many of these studies utilize recurrent DL models, which excel at processing temporal data effectively [3].

In this paper, deep learning models are built using the Convolutional Neural Network and Recurrent Neural Network i.e CNN, LSTM, and hybrid CNN-LSTM and estimation of SOC and performance analysis of these models are carried out. SOC estimation error is the main performance criterion. This project helps in identifying the strengths and weaknesses of each model. This method is data driven which estimates SOC with less computational complexity, higher accuracy and adaptability than other conventional equivalent circuit and electrochemical models. And Better SOC prediction ensures optimal energy usage and reliable range prediction for a vehicle

II. LITERATURE REVIEW

Several researches have been conducted in this field recently. Some of them are traditional architectures that are no longer up to date and are slow and time consuming.

Chemali et al.[4] explores the use of Recurrent Neural Network(RNN) with Long Short-Term Memory (LSTM) networks for estimating the State of Charge (SOC) of lithium-ion batteries. RNN-LSTM based approach learns directly from voltage, current, and temperature data, eliminating the need for complex battery models, filters or inference systems like Kalman Filters. The estimated SOC achieves a low Mean Absolute Error of 0.00573 at a fixed ambient temperature and a MAE of 0.01606 on a dataset with ambient temperature increasing from 10 °C to 25°C

Xiangbao et al.[5] proposed CNN-LSTM network to infer battery SOC from the voltage, current, and temperature. The literature shares that CNN-LSTM network well captures the nonlinear relationship between SOC and measurable variables and shows that it has better performance than individual LSTM and CNN networks. In this paper 1D Convolutional layer with six filters of length three is adopted to capture spatial features of battery variables, and one LSTM layer with 300 hidden nodes is added to learn the relationship between current SOC and historical inputs.

Fangfang et al.[6] used LSTM to model the complex dynamics of lithium iron phosphate batteries and infer battery SOC from measurable variables. The LSTM architecture

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starts with a sequence of three input vectors followed by three hidden layers with 50 nodes, then a fully connected layer is added later to scale and transform the output further. The proposed network presents quick convergence of SOC, with 2% MSE and MAE of 1

Hannan et al.[3] gave a comparative performance analysis of CNN, GRU, LSTM, and FCN where CNN model consists of a single temporal convolution layer with 22 filters and a kernel size of 5 followed by a max pooling layer with pool size 2. The number of hidden units and filters were selected such that all models consist of approximately the same number of parameters for a fair comparison. The proposed model FCA outperformed conventional DL models such as the LSTM, GRU, and CNN by scoring the lowest RMSE, and MAE and CNN performed worst among these models.

Fujin et al.[7] proposed a Physics-Informed Neural Network (PINN) for lithium-ion battery State of Health (SOH) estimation. The PINN integrates empirical degradation models and state space equations with neural networks, enabling it to model battery degradation dynamics accurately and stably. The authors introduce a feature extraction method that uses data from the constant voltage charging stage, ensuring the approach's applicability to different battery chemistries and charge/discharge protocols. The performance of the method was compared with the other Neural Networks like CNN, MLPS and the proposed PINN demonstrated superior performance in regular, small sample, and transfer experiments, achieving a mean absolute percentage error (MAPE) of 0.87%, outperforming alternative neural networks. The study highlights the potential of hybrid physics-based and data-driven methods for battery SOH estimation and degradation modeling.

III. MATERIALS AND METHODS

Initially during the data preprocessing stage null values present in the dataset were dropped and data was normalized to a fixed range i.e [0, 1] to make it suitable to implement on deep learning models. So, after data preprocessing different model architectures were designed based on the deep learning framework and evaluated based on the accuracy. The loss function chosen for these models is mean absolute percentage error (MAPE).

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100$$

The hyperparameters chosen for these models was varied to obtain the parameters with the least MAPE for each of the three models. These hyperparameters include the number of epochs, batch size, learning rate, and sequence length. Adam optimizer was chosen with a different learning rate for different models.

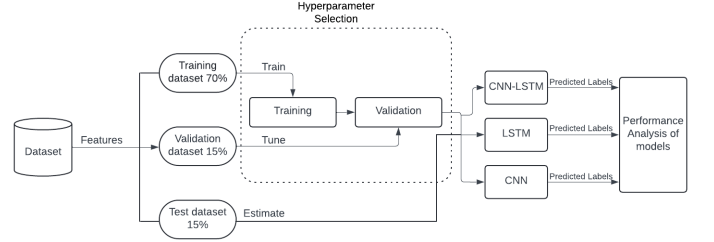


Fig. 1: flow diagram

A. Data Description

The dataset consists of 81138 data points, including measurement of voltage, current, and timestamps, along with the corresponding State of Charge (SOC) values. Timestamps indicating when the data was collected, essential for understanding temporal patterns in battery behavior. Voltage measured continuously influences SOC levels, likewise continuous current measurement is crucial for assessing charge and discharge cycles. The target variable representing the remaining charge in the battery i.e SOC, typically expressed as a percentage.

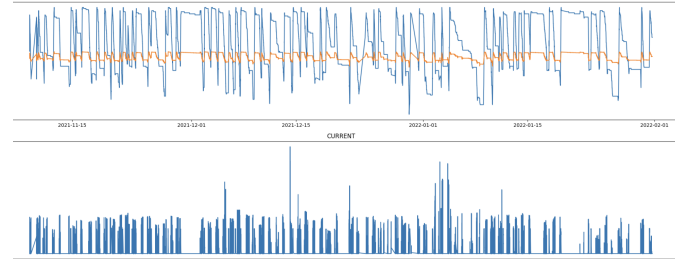


Fig. 2: Variation of Voltage, Current and SOC with respect to time

B. Model Architecture

1) *CNN*: The model takes input data with three channels corresponding to the features: voltage, current, and timestamp. The architecture employs five 1D Convolutional layers each with kernel size of three and padding and stride of one. For the given input, Convolution layer applies a sliding filter of size three to input sequence. This layer outputs feature maps where each map highlights specific patterns and then normalizes the feature maps by rescaling and shifting. Batch normalization is used in every layer to improve convergence and stabilize training. ReLU is used as an activation function to introduce non-linearity to help models to learn complex relationships.

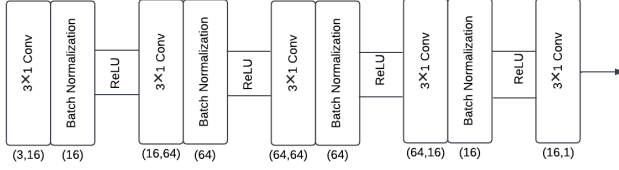


Fig. 3: CNN Architecture

2) *LSTM*: The model takes 3 neurons as input and has 32 hidden nodes in the LSTM cell. Thus the model learns 32 different features from the input sequence. The model uses 2 stacked LSTM layers, helps to learn complex temporal dependencies. The hidden states are passed through a dense layer to make the final prediction. A simple linear transformation maps the hidden state to the target output which is the predicted SOC[8].

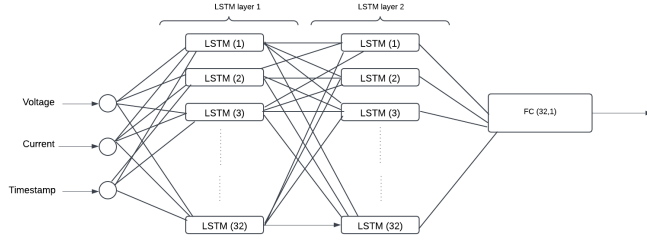


Fig. 4: LSTM Architecture

3) *CNN-LSTM*: This hybrid model has three 1D Convolutional layers where first Convolution layer extracts low-level features from the input sequence using filter size 3 and outputs into 64 features maps, second layer builds on the low-level features, learning higher-level representations, and the third layer reduces the feature maps back to 64 channels, condensing learned representations. ReLU as activation function and output of CNN is fed into the LSTM layer which has 2 layers having each 32 nodes. A single value predicted SOC is obtained per sequence from the final hidden layer using simple linear transformation.

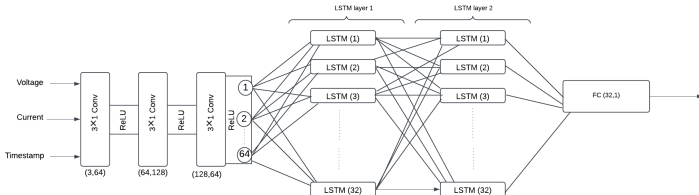


Fig. 5: CNN-LSTM Architecture

IV. EXPERIMENTAL RESULTS

All three models were trained with different hyperparameters to get the best result possible. Given table represents all the parameters used.

TABLE I: Initialized Hyperparameters

Model	Learning Rate	Epochs	Batch Size	Weight Decay
CNN	0.0005	80	60	0.0002
LSTM	0.0001	100	60	0.0002
CNN-LSTM	0.0005	100	60	0.0002

A. CNN Result

The training and validation accuracy observed for CNN after are 0.5405 and 0.3061 respectively. The parameters were chosen according to the best training result. For CNN, 80 epochs were initialized but early stopping was triggered at 59 epochs and the final training Loss and validation loss were 0.014 and 0.047. The large fluctuations in the validation accuracy indicates instability. Training loss shows a steady decline indicating that the model is successfully reducing error on the training data. Unlike the training loss, the validation loss fluctuates significantly and starts increasing after some epochs. The increasing trend in validation loss indicates that the model's performance on this data deteriorates as training progresses.

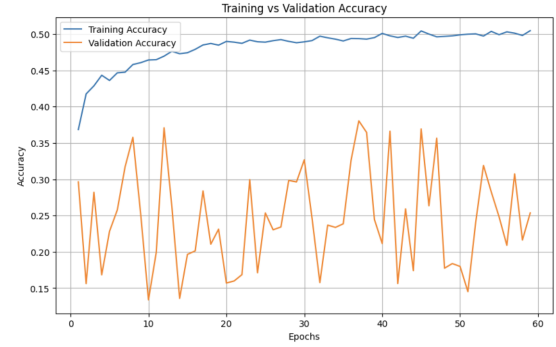


Fig. 6: Accuracy plot of CNN

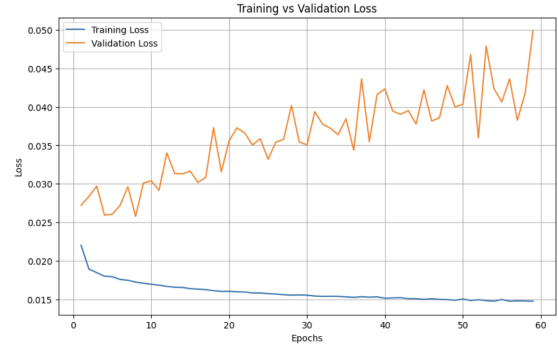


Fig. 7: Loss plot of CNN

B. RNN Result

The model was trained up to 100 epochs and stabilizes around 90% training accuracy and around 80% validation accuracy after approximately 20 epochs. Both training loss and validation loss decrease steeply in the first few epochs

and then stabilize at very low values. The decrease in loss indicates that the model is effectively optimizing its parameters to minimize the error on both training and validation sets. The low values of both indicates quick convergence and reflects the effectiveness in learning the patterns in the data and making accurate predictions.

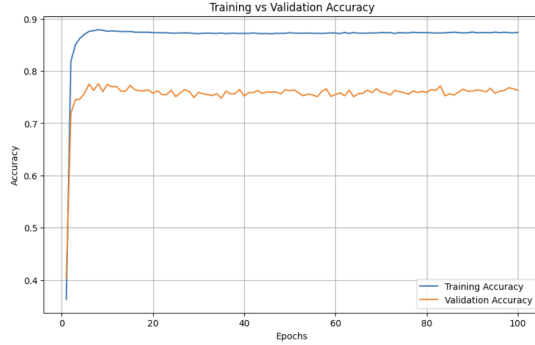


Fig. 8: Accuracy plot of RNN

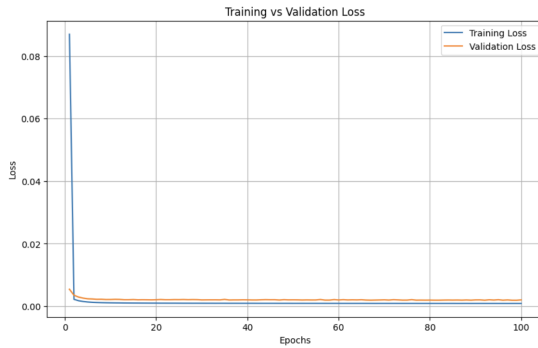


Fig. 9: Loss plot of RNN

C. CNN-LSTM Result

The training accuracy for the hybrid model is around 90% and the validation accuracy is around 80%. Early stopping was triggered at 76 epochs and the validation accuracy fluctuates significantly, with noticeable peaks and troughs throughout training. These fluctuations indicate sensitivity to data points during validation and is less stable indicating that the CNN-LSTM model is facing generalization issues.

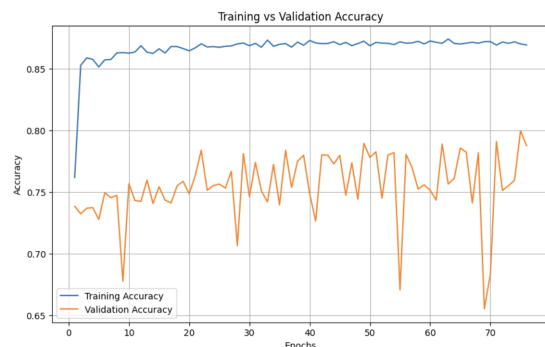


Fig. 10: Accuracy plot of CNN-RNN

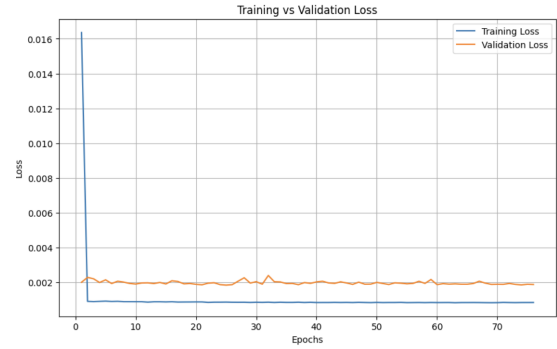


Fig. 11: Loss plot of CNN-RNN

TABLE II: Training Results of all three models

Metric	CNN	LSTM	CNN-LSTM
Training Accuracy	54.08%	87.57%	88.21%
Validation Accuracy	30.61%	76.31%	76.31%
Training Loss	1.45%	0.09%	0.08%
Validation Loss	4.78%	0.20%	0.19%
Test Accuracy	31.50%	74.98%	82.55%
Test Loss	8.60%	0.11%	0.1%

V. CONCLUSION

In this project each model was trained using different numbers of layers, different activation functions like ReLU, and Leaky ReLU, different hyperparameters. In each epoch, CNN took the least time, followed by LSTM and hybrid model. Because LSTM and CNN-LSTM perform multiple recurrent calculations in the time direction, while CNN only does it once. Similarly, CNN converged the fastest, within 60 epochs. LSTM has the slowest convergence and took about 100 epochs, so its total training time is more than that of both CNN and CNN-LSTM.

It is noted that CNN gave accuracy around 50% only, LSTM around 90% and hybrid model also around 90%. Based on the performance, CNN struggled capturing temporal dependencies and has limited generalizability. Thus, performed worst among three models. The LSTM model which is designed to handle sequential data achieved stable training and validation accuracy showcasing its ability to capture temporal relationships for SOC estimation. The hybrid model leveraged with both spatial and temporal features, achieved high accuracy among these models but validation accuracy exhibited significant fluctuations. Among the three models, the LSTM model emerges as the most reliable for SOC estimation, balancing stable performance with effective generalization, making it better suited for real-world applications.

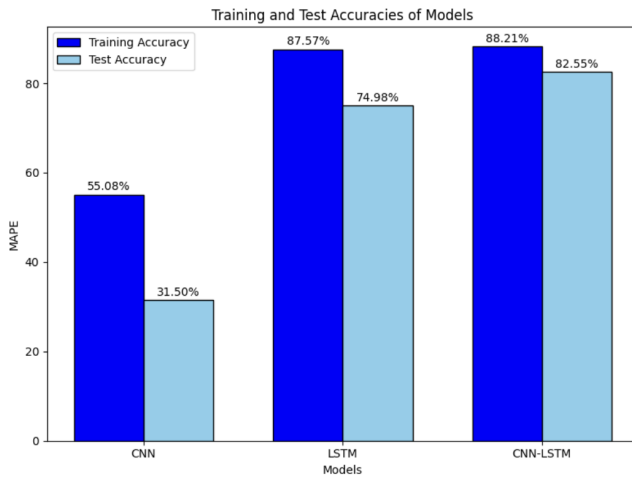


Fig. 12: Performance graph of models

Future Work

With the accuracy that is achieved in hybrid models, it is not the optimum performance. To get the best result model can be experimented with a diverse dataset including features like Temperature, aging effect and using advanced regularization techniques. Transfer learning can also be implemented to get the best result from the model where the model is pre-trained on a large, generic battery data and fine tuning it to estimate on specific SOC.

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